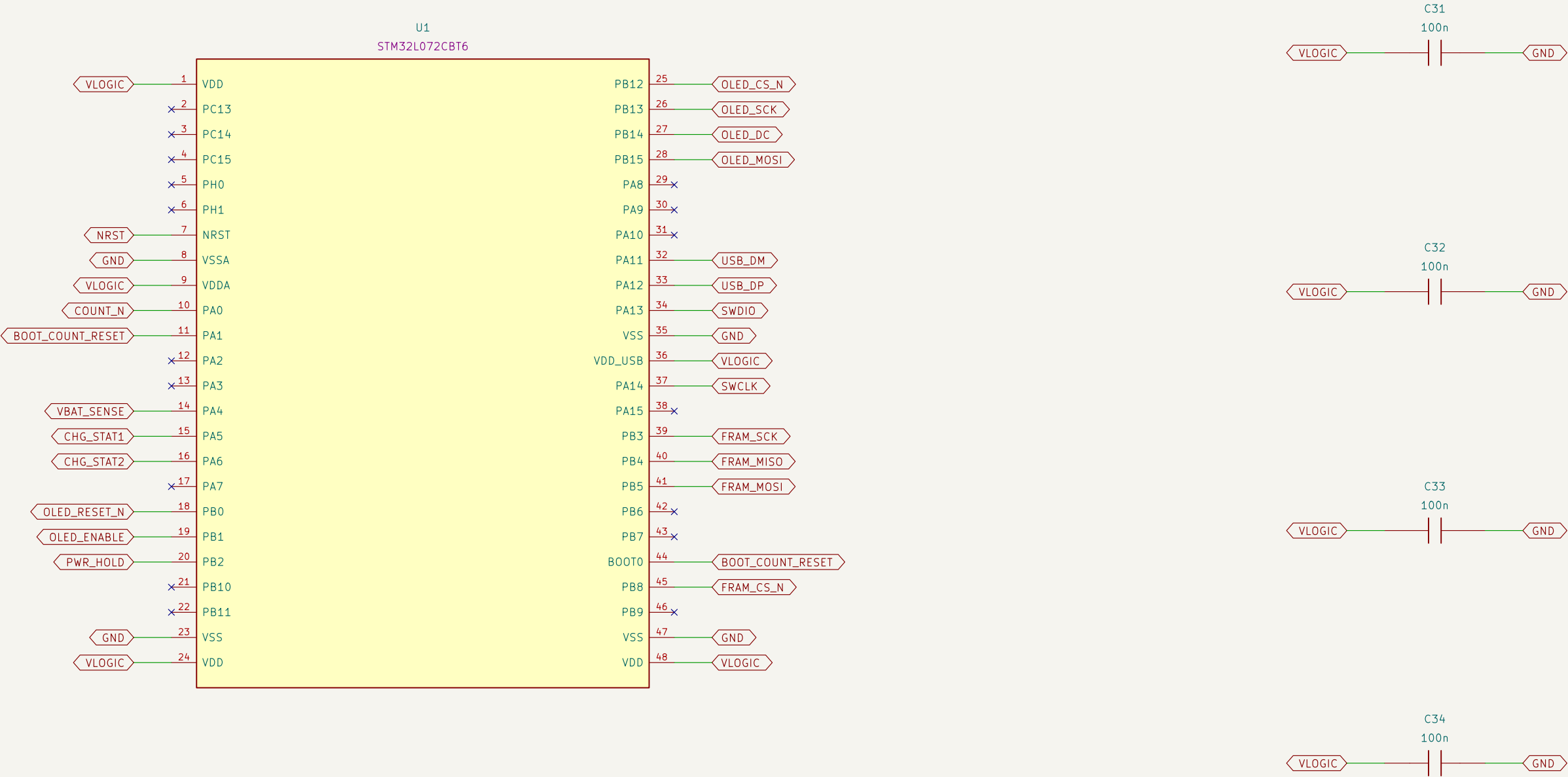
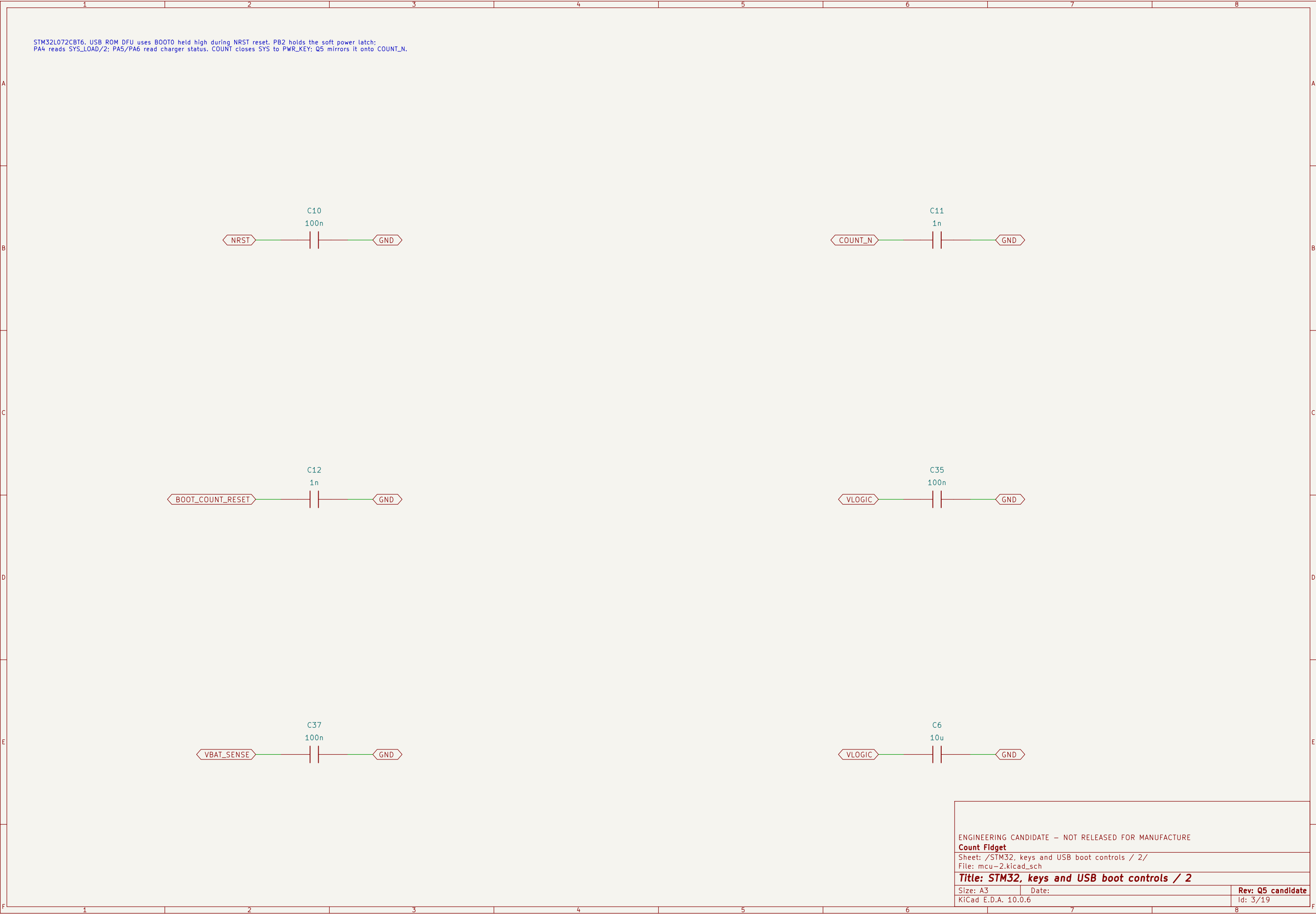
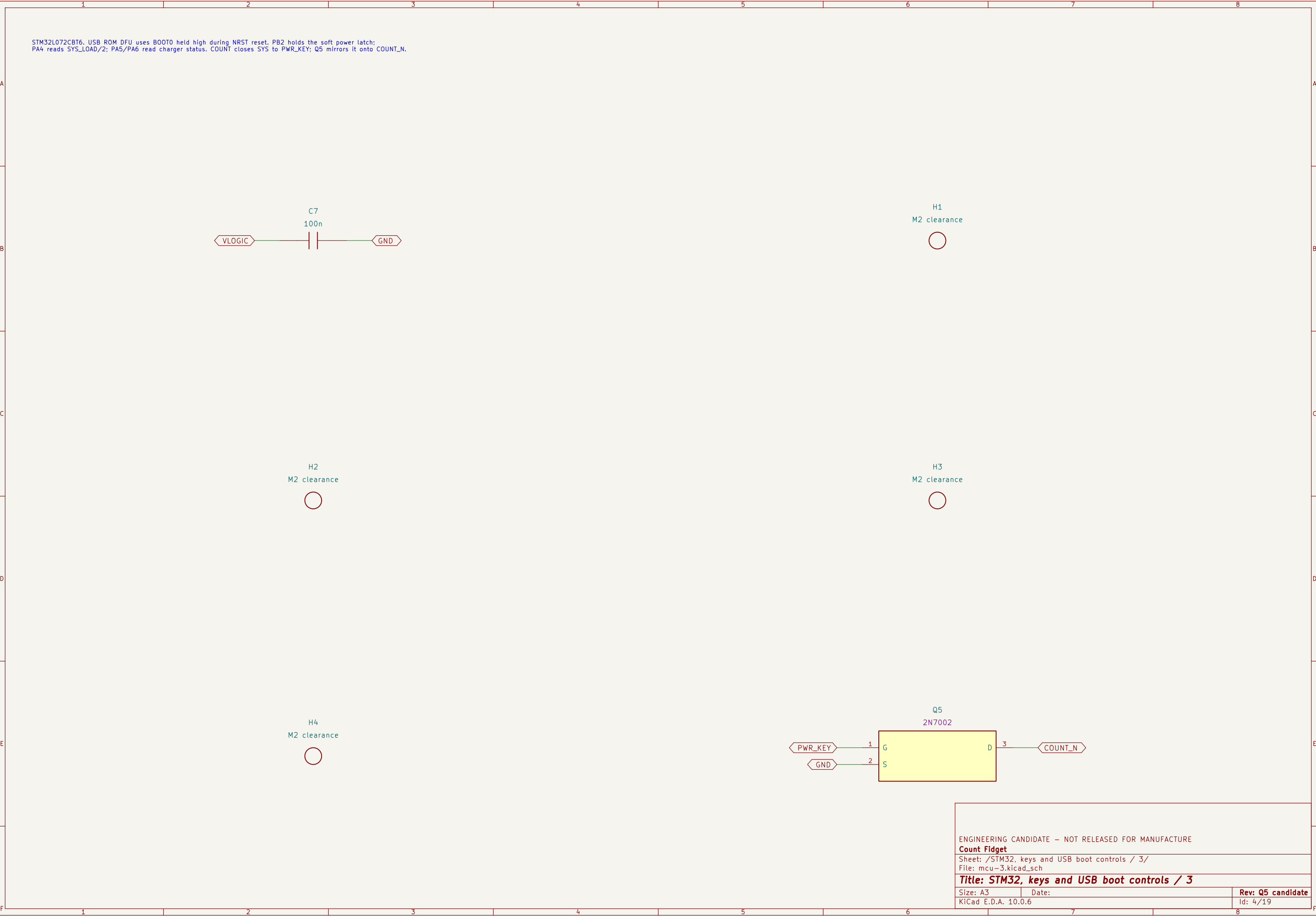


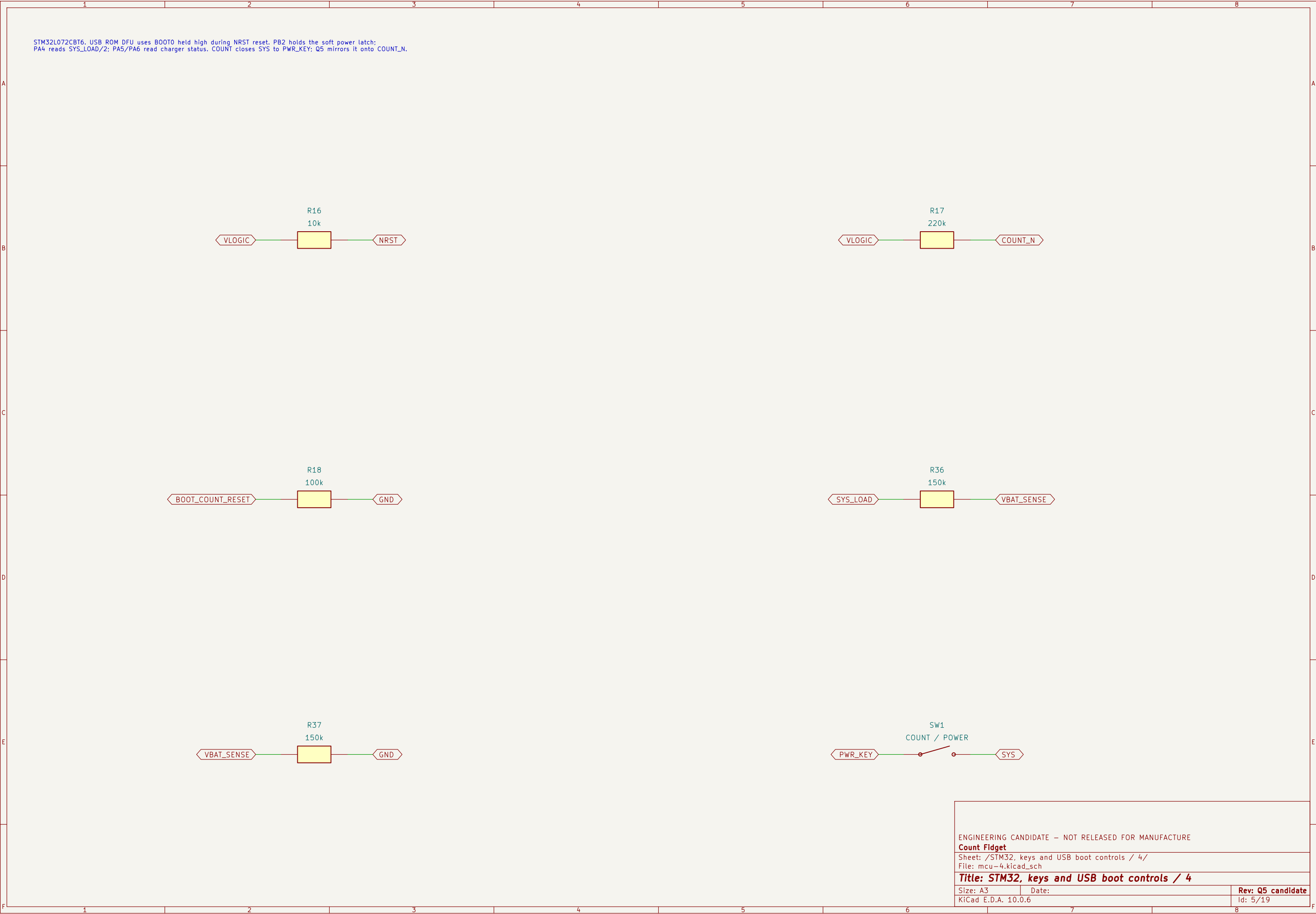
	1	2	3	4	5	6	7	8	
A	Q5 schematic index – 81 fitted PCB parts plus non-fitted test lands and mounts. Global net labels connect the functional sheets. Crosses explicitly mark unused physical pins. Q1 submitted files remain frozen. This candidate still requires ERC, layout, first-article and safety qualification.								A
	STM32, keys and USB boot controls		STM32, keys and USB boot controls / 2			STM32, keys and USB boot controls / 3			
	U1 / C31 / C32 / C33 / C34		C10 / C11 / C12 / C35 / C37 / C6			C7 / H1 / H2 / H3 / H4 / Q5			
B	File: mcu-1.kicad_sch STM32, keys and USB boot controls / 4		File: mcu-2.kicad_sch STM32, keys and USB boot controls / 5			File: mcu-3.kicad_sch STM32, keys and USB boot controls / 6			B
	R16 / R17 / R18 / R36 / R37 / SW1		SW2 / SW3 / TP1 / TP11 / TP12 / TP2			TP3 / TP4 / TP5 / TP6 / TP9			
	File: mcu-4.kicad_sch SPI OLED module and power control / 1		File: mcu-5.kicad_sch SPI OLED module and power control / 2			File: mcu-6.kicad_sch Independent SPI retained-count memory / 1			
C	C20 / C28 / D51 / R24 / R25 / R29		U6			C30 / C36 / R32 / R33 / U8			C
	File: display-1.kicad_sch USB, charging, soft power latch and common supply / 1		File: display-2.kicad_sch USB, charging, soft power latch and common supply / 2			File: memory-1.kicad_sch USB, charging, soft power latch and common supply / 3			
	C1 / C13 / C2 / C29 / C3 / C38		C39 / C40 / C5 / D1 / D2 / D3			D4 / J1 / R1 / R19 / R2 / R3			
D	File: usb-power-1.kicad_sch USB, charging, soft power latch and common supply / 4		File: usb-power-2.kicad_sch USB, charging, soft power latch and common supply / 5			File: usb-power-3.kicad_sch On-board LIR2032 holder and protection / 1			D
	R30 / R31 / R34 / R35 / R38 / R4		R5 / R6 / TP13 / U2 / U3 / U7			BT1 / C4 / Q1 / Q2 / R7 / R8			
	File: usb-power-4.kicad_sch On-board LIR2032 holder and protection / 2		File: usb-power-5.kicad_sch Independent charge-temperature window / 1			File: battery-1.kicad_sch Independent charge-temperature window / 2			
E	R9 / U4		C9 / Q3 / Q4 / R10 / R11 / R12			R13 / R14 / R15 / TH1 / U5			E
	File: battery-2.kicad_sch		File: thermal-1.kicad_sch			File: thermal-2.kicad_sch			
F	1	2	3	4	5	6	ENGINEERING CANDIDATE – NOT RELEASED FOR MANUFACTURE Count Fidget Sheet: / File: click-counter-Q5.kicad_sch Title: Count Fidget – Q5 engineering candidate Size: A3Date:KiCad E.D.A. 10.0.6 Rev: Q5 candidateId: 1/19	8	F

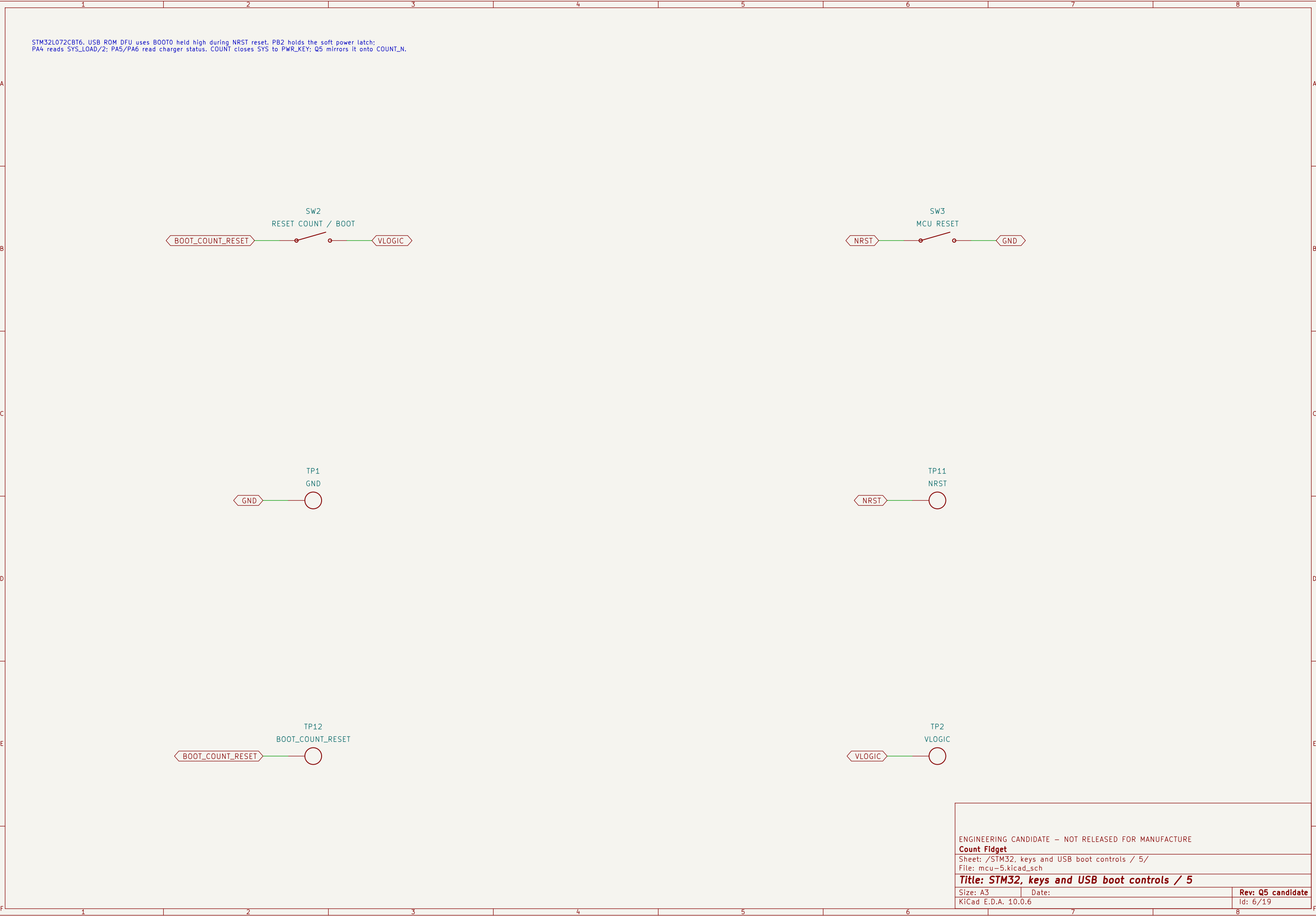
STM32L072CBT6. USB ROM DFU uses BOOT0 held high during NRST reset. PB2 holds the soft power latch; PA4 reads SYS_LOAD/2; PA5/PA6 read charger status. COUNT closes SYS to PWR_KEY; Q5 mirrors it onto COUNT_N.

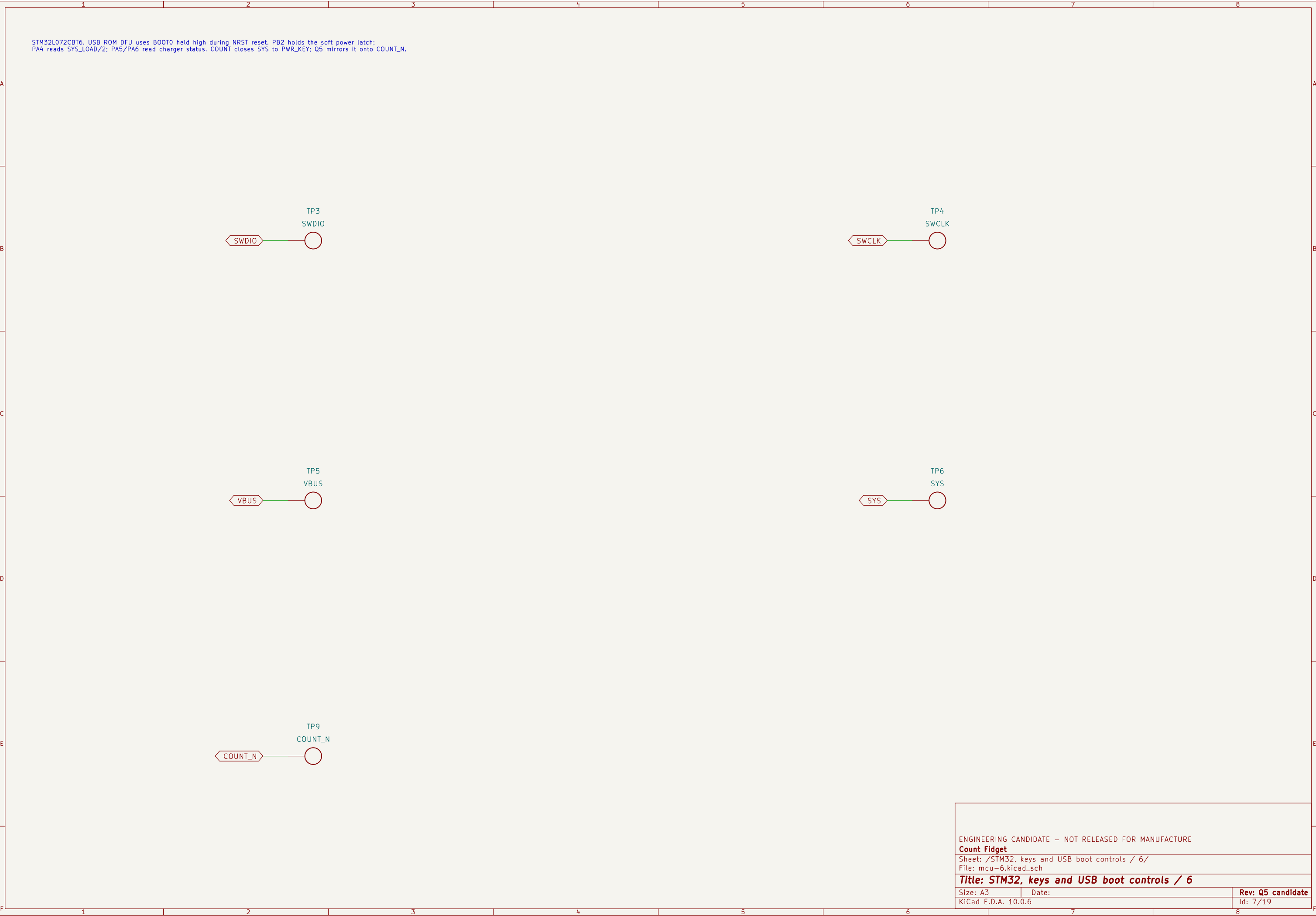












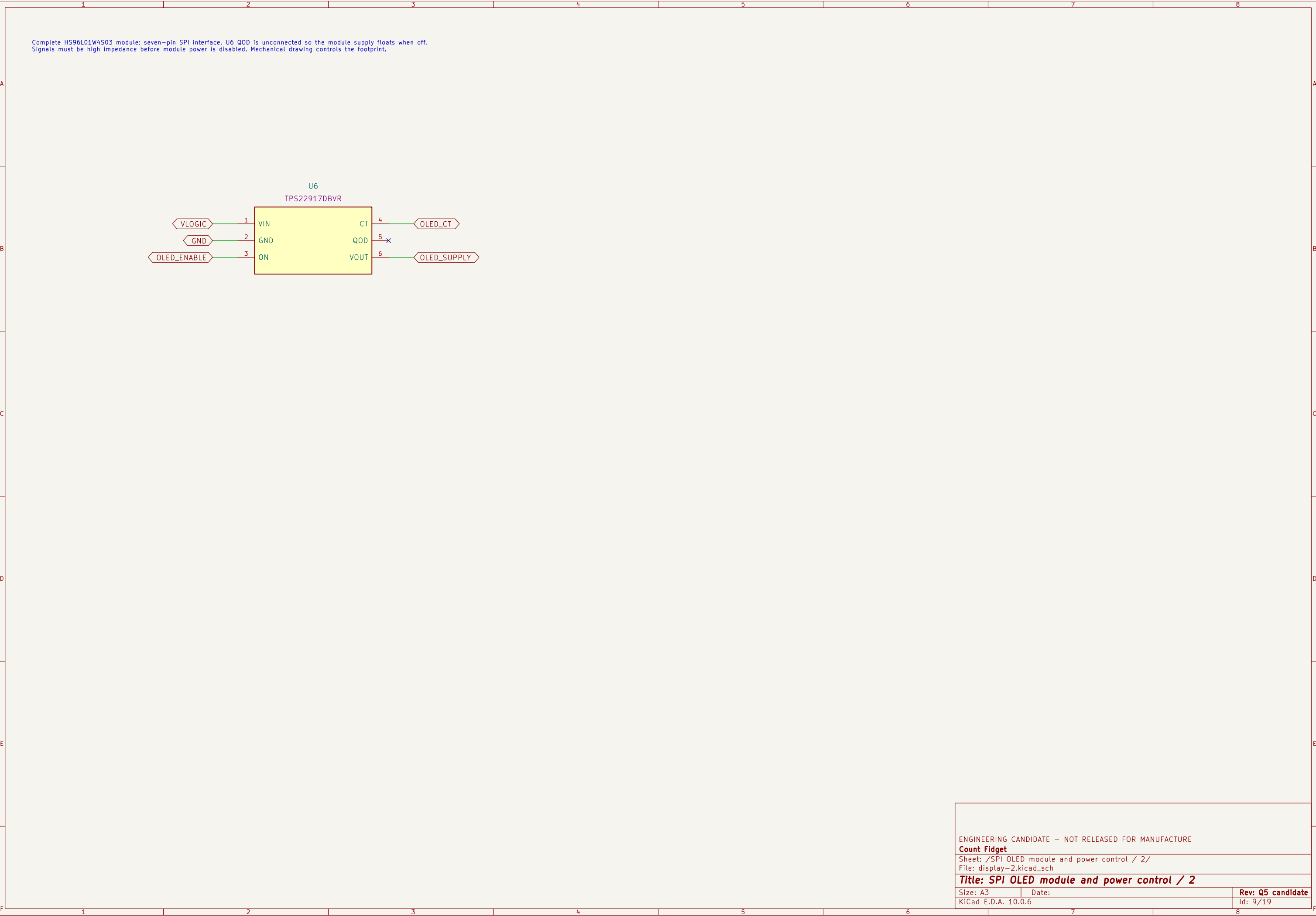
Complete HS96L01W4S03 module; seven-pin SPI interface. U6 QOD is unconnected so the module supply floats when off. Signals must be high impedance before module power is disabled. Mechanical drawing controls the footprint.

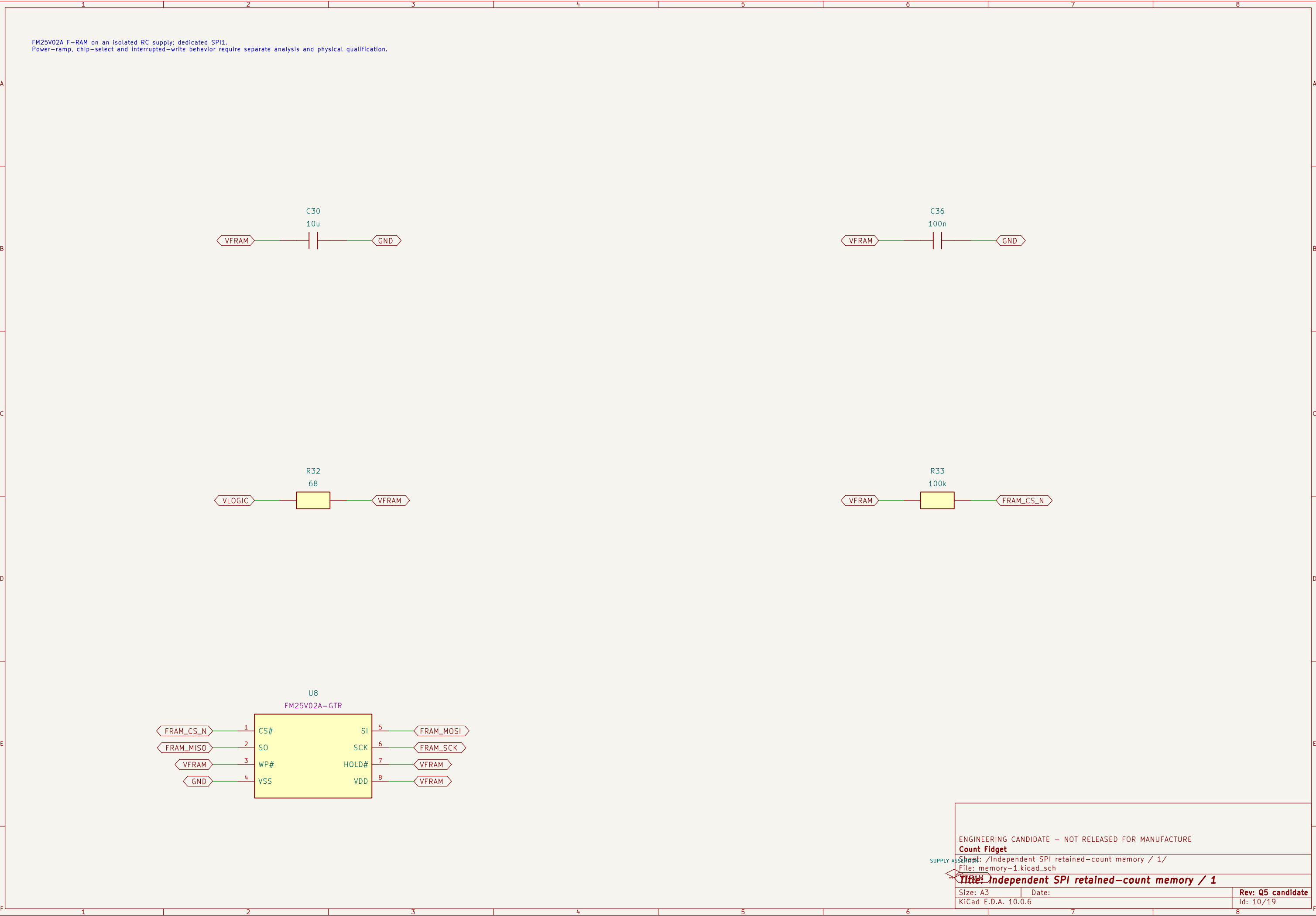
The schematic diagram illustrates the electrical connections for an SPI OLED module and its power control. The central component is the HS96L01W4S03 module (DS1), which is connected to various signals and power sources. The connections are as follows:

- Power and Ground:**
 - GND:** Connected to pin 1 of the module.
 - OLED_SUPPLY:** Connected to pin 2 of the module.
 - OLED_RESET_N:** Connected to pin 5 of the module.
 - OLED_DC:** Connected to pin 6 of the module.
 - OLED_CS_N:** Connected to pin 7 of the module.
- Signal Connections:**
 - OLED_SCK:** Connected to pin 3 of the module.
 - OLED_MOSI:** Connected to pin 4 of the module.
 - RES#:** Connected to pin 5 of the module.
 - D/C#:** Connected to pin 6 of the module.
 - CS#:** Connected to pin 7 of the module.
- Capacitors:**
 - C20 (100nF):** Connected between VLOGIC and GND.
 - C28 (4.7nF):** Connected between VLOGIC and OLED_CT.
- Resistors:**
 - R24 (100k):** Connected between OLED_ENABLE and GND.
 - R25 (100k):** Connected between OLED_RESET_N and GND.
 - R29 (100k):** Connected between OLED_SUPPLY and OLED_CS_N.

The diagram also includes a title block with the following information:

- ENGINEERING CANDIDATE – NOT RELEASED FOR MANUFACTURE
- Count Fidget
- Sheet: /SPI OLED module and power control / 1/
- File: display-1.kicad_sch
- Title: SPI OLED module and power control / 1**
- Size: A3 | Date: | Rev: Q5 candidate
- KiCad E.D.A. 10.0.6 | Id: 8/19





U7 ON = VBUS (D4) OR COUNT key (D3 A1) OR PWR_HOLD (D3 A2), pulled down by R35.
Off: only BQ29700 and BQ25185 battery quiescent current remain. Cold cell insertion is not proven by the slew switch.

Components and Connections:

- C1:** 2.2uF capacitor connected between VBUS and GND.
- C2:** 10uF capacitor connected between SYS and GND.
- C3:** 2.2uF capacitor connected between CELL_P and GND.
- C13:** 4.7nF capacitor connected between RC_COMP and GND.
- C29:** 4.7nF capacitor connected between SYS and SYS_CT.
- C38:** 100nF capacitor connected between VBUS and GND.
- R35:** Resistor connected between VBUS and GND.
- Supply Assertions:** VBUS and GND are marked as SUPPLY ASSERTION.

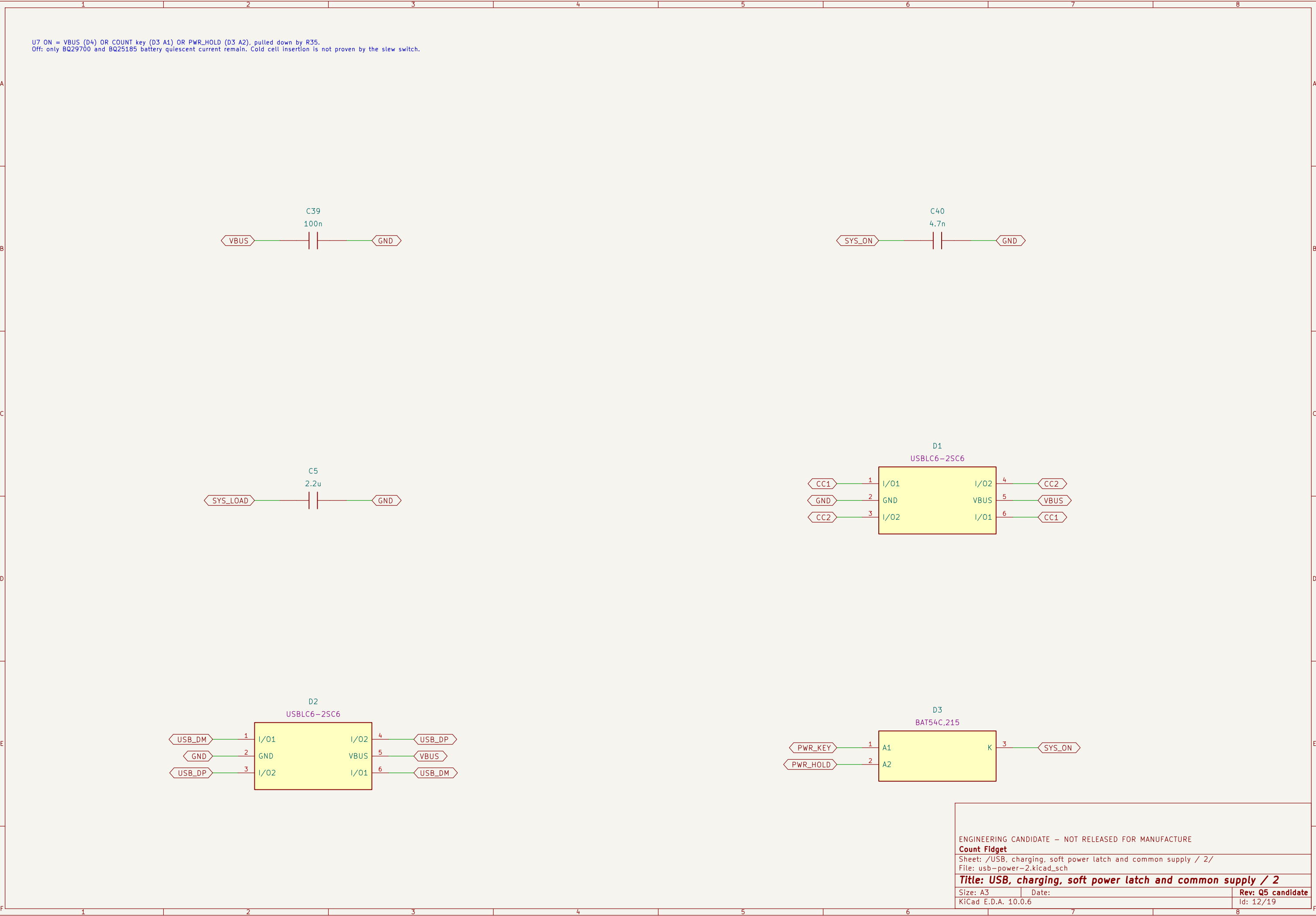
Title Block:

ENGINEERING CANDIDATE – NOT RELEASED FOR MANUFACTURE		
Count Fidget		
Sheet: /USB, charging, soft power latch and common supply / 1/ File: usb-power-1.kicad_sch		
Title: USB, charging, soft power latch and common supply / 1		
Size: A3	Date:	Rev: Q5 candidate
KiCad E.D.A. 10.0.6		Id: 11/19

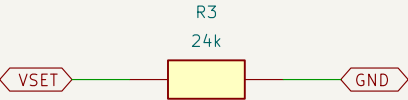
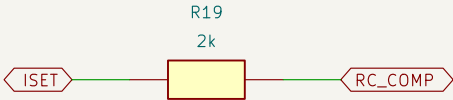
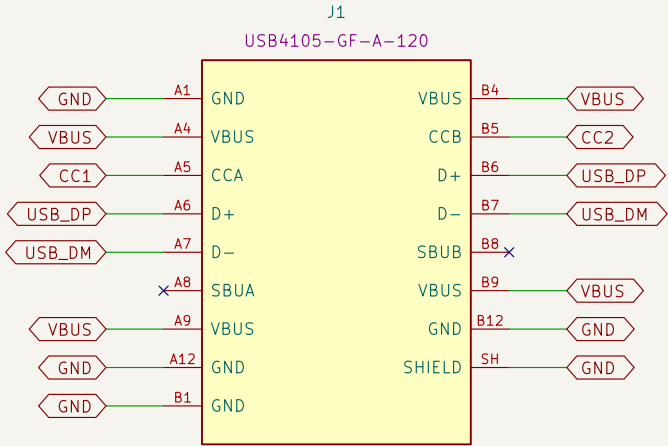
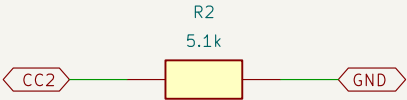
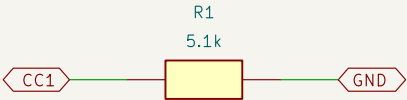
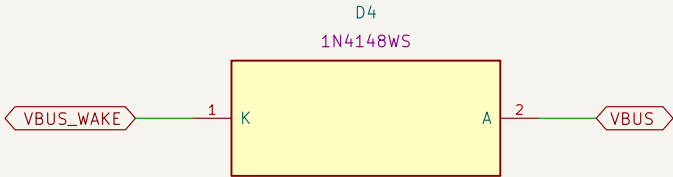
Count Fidget

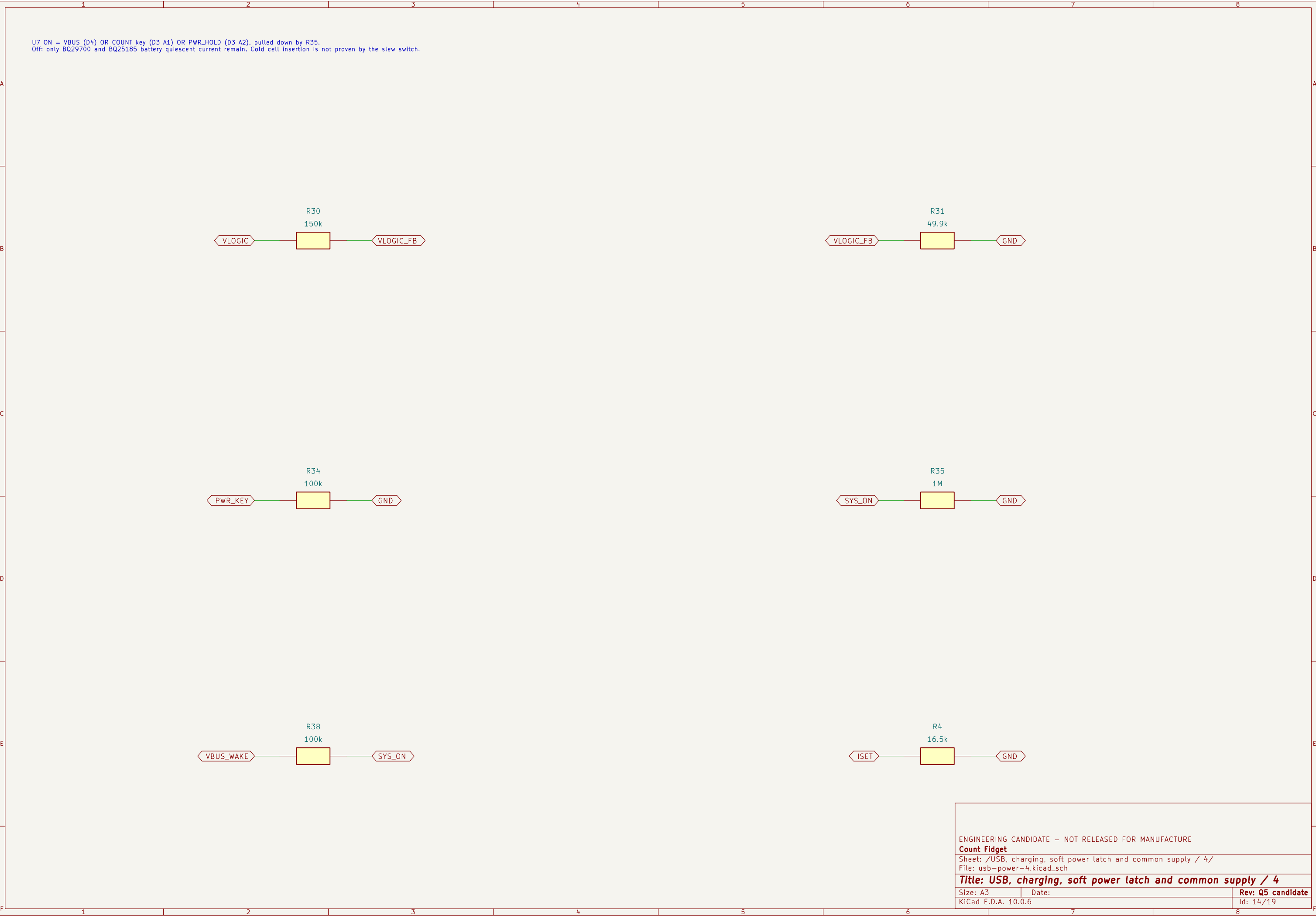
Title: USB, charging, soft power latch and common supply / 1

Size: A3	Date:	Rev: Q5 candidate
KiCad E.D.A. 10.0.6		Id: 11/19

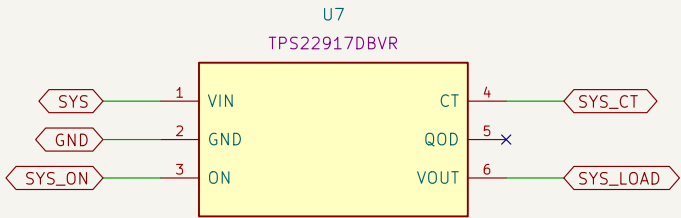
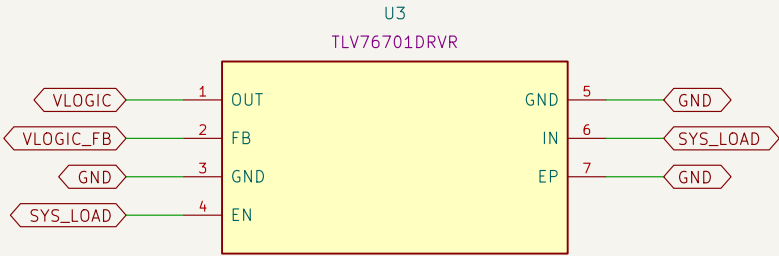
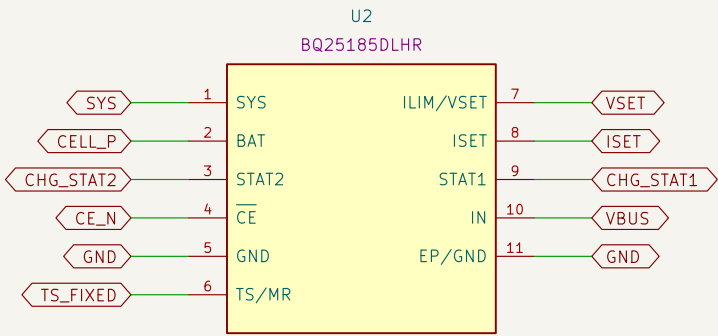
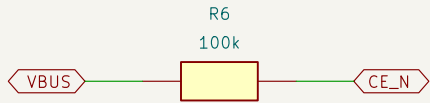
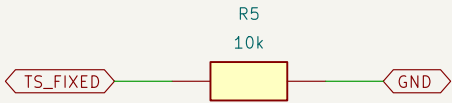


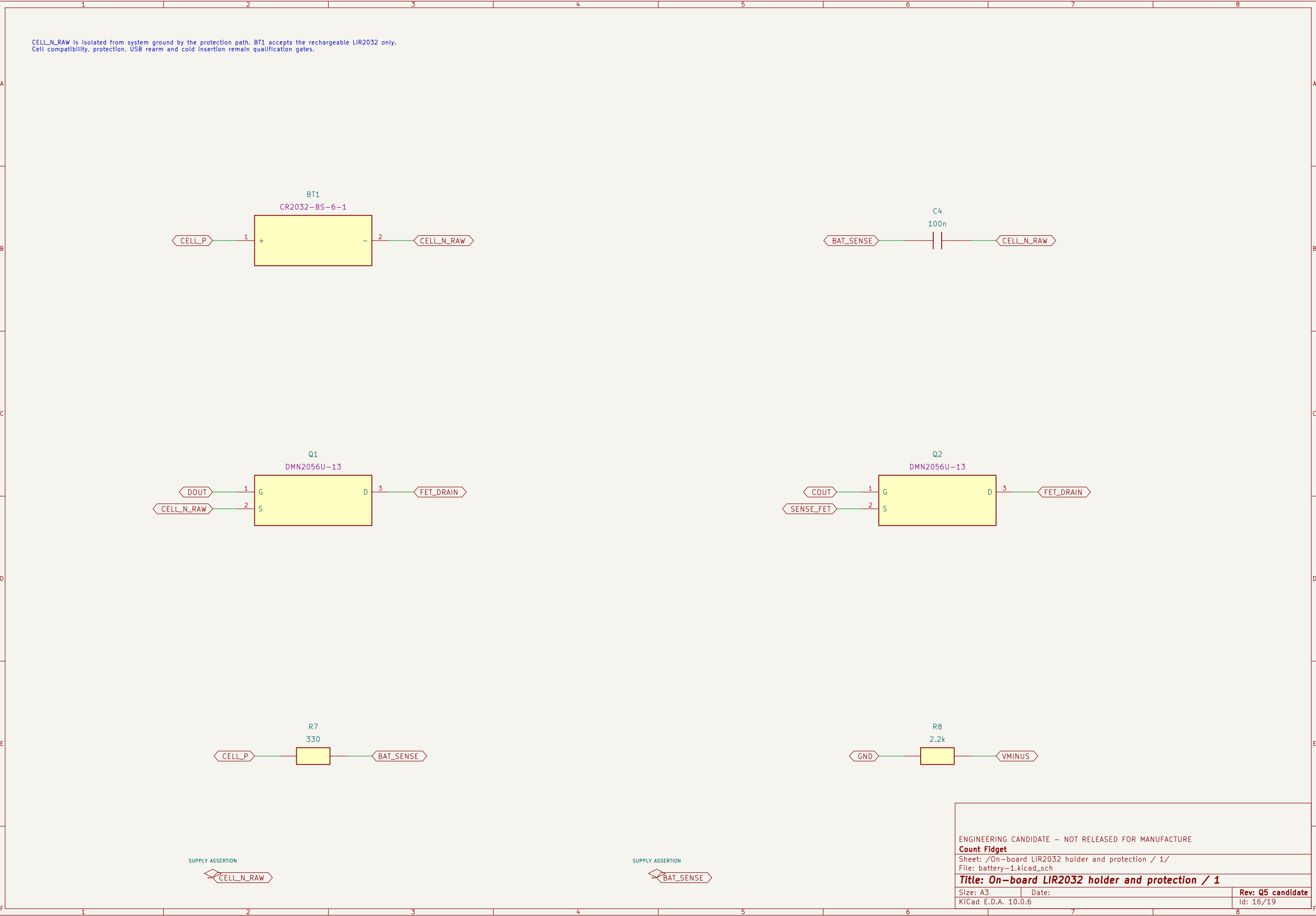
U7_ON = VBUS (D4) OR COUNT key (D3 A1) OR PWR_HOLD (D3 A2), pulled down by R35.
Off: only BQ29700 and BQ25185 battery quiescent current remain. Cold cell insertion is not proven by the slew switch.

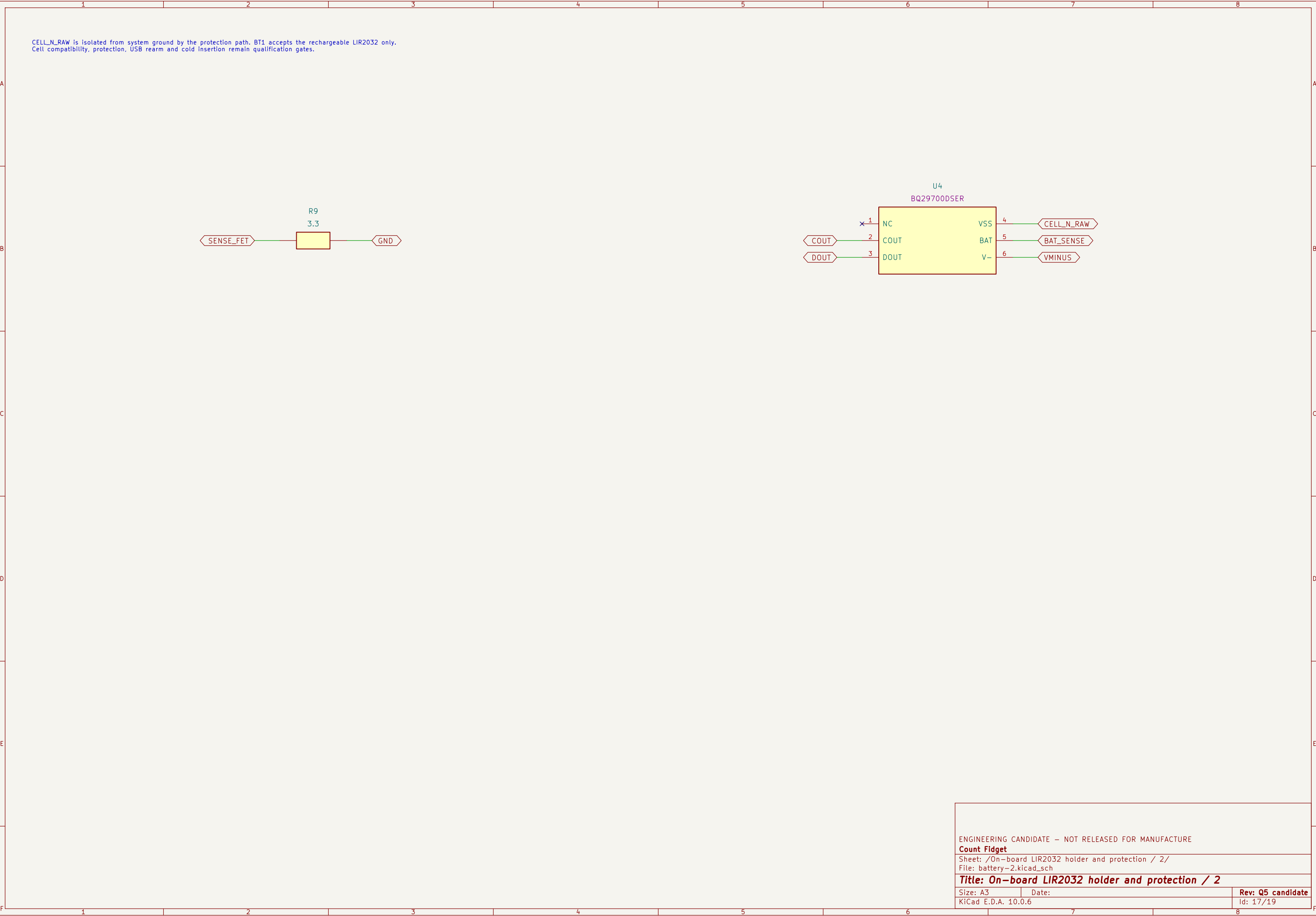




U7 ON = VBUS (D4) OR COUNT key (D3 A1) OR PWR_HOLD (D3 A2), pulled down by R35.
Off: only BQ29700 and BQ25185 battery quiescent current remain. Cold cell insertion is not proven by the slew switch.







TH1 on the board beside BT1; both window outputs must enable their series FETs for charging.
Gate pulldowns disable charging when comparator power is absent; threshold and fault tests remain required.

